Triggers for irrigation decision-making in greenhouse horticulture using Internet of Things

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***Abstract*—In this paper, an electrical conductivity (EC) sensor is designed specifically to measure drainage from greenhouse horticulture. The EC measured value gives information about minerals content in the input and output (drainage) from different products. Furthermore, a Fuzzy Logic Controller (FLC) with IoT capabilities is developed for analysis and linguistic decision making about fertigation (fertilizers + water) in a greenhouse. The result is a controller that embeds structural human knowledge about irrigation scheduling into an analytic FLC, in order to provide agricultural scientists with quick access to a particular crop's main production and growth related variables, and allowing for future data driven decisions on the spot. Final results support the continuing development of affordable yet efficient sensors and applications based on FLC and IoT, that provides effective production tools for horticulture producers in developing countries.**

***Index Terms*—Agriculture in protected environments, Electrical conductivity sensor, Fuzzy Logic Controller, greenhouse horticulture, Internet of Things.**

I. Introduction

Nowadays, researchers involved with decision-making about irrigation for the cultivation of vegetables without soil, have to collect and measure manually quantity of drainage and electrical conductivity (EC) of this drainage, in order to determine the level of utilization of the nutrients supplied by the fertigation system. Is important to mention that the manual method requires the daily measurement of the amount and EC of drainage. In figure 1 is shown a standalone controller with IoT capabilities, automatic measurement of drainage, pH, EC, ambient temperature and humidity. This paper proposes a new method to improve the one established in [1]. Specifically, the aim is to solve the problem related to the continuous measurement of EC and how this variable is defined as a linguistic trigger for decision-making, it also mentions how this EC sensor was tested and validated before applying for a patent that takes in consideration the capacity of the EC sensor to remain submerged in the preparation and storage tank of nutrients that will be supplied to the crops through fertigation. The EC sensor is considered to be resistant to highly saline aqueous media and can stay submerged in this media for long periods (months), without calibration or maintenance.

The coconut fiber substrates used in hydroponic cultivation are composed of an inert material, therefore all the nutritional requirements of the plants must be supplied through fertigation, both the nutrient solution and drainage are highly saline and therefore the sensor to determine the EC must be resistant to said aqueous medium. The sensor is used to measure the EC of drainage in an automated way and with interconnection to an Internet of Things platform. More details of the mentioned EC sensor, are in [2], [3], [4].

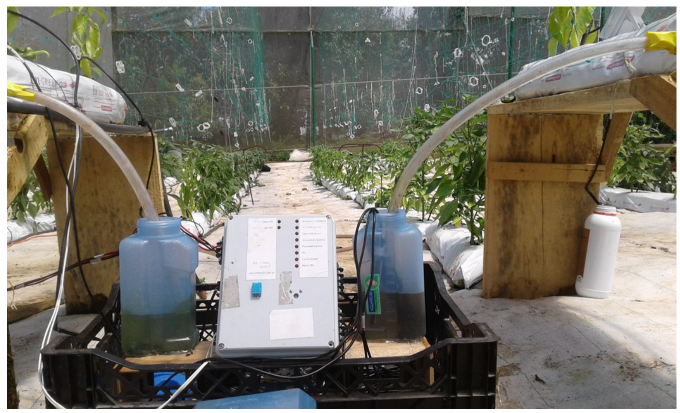

Figure 1. Standalone controller with IoT capabilities, automatic measurement of drainage, pH, EC, ambient temperature and humidity. Collected data is send to ThingSpeak IoT platform [3].

This paper is organized as follows: Section II describes the problems regarding irrigation decision-making considering linguistic terms related to water availabitilty measuring drainage, EC and potential of hydrogen (pH). Section III presents the general structure of the electronic data collecting system, along with the application specific software layers needed. Some explanations are given on the IoT software platform being used and how it is integrated to the acquisition and processing system. In section IV an standalone FLC application in greenhouse horticulture is described. Finally, some conclusions and perspectives are given in Section V.

II. Problem Description

In general, the sensors used to determine the EC in aqueous media have the disadvantage that they cannot remain submerged for long periods of time since they are explicitly designed to instantly measure and display the measured data

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and the person responsible of recording these data must do it in site, it means directly and manually in the greenhouse measuring each container quantity of drainage, EC, and pH (see again figure 1, plastic container under wood structure).

In [5] a Fuzzy Logic Controller (FLC) with two inputs is analyzed, in this paper a new method including a third input variable and an embedded FLC is developed. Is important to keep in mind that water is considered as the main input for greenhouse horticulture, agriculture and food production, because all variables related to water are affected by the excess or deficit of rainfall, changes in temperature and other associated environmental variations, which may be particularly severe in fragile tropical ecosystems.

A. Test of probes for EC sensor

Considering the problem mentioned above, several materials were analyzed in order to find one with the following capabilities: affordable, durable, low or none maintenance and of course, a precision similar or better than commercial EC sensors.

In figure 2 are shown several probes after their being tested in a salt spray chamber. An experiment was designed with 24 probes, every 24 hours 4 probes were taken out of the chamber, specifically test time were set to 24, 48, 72, 120, 144 and 168 hours (See Table I).

Table I
CURRENT DATA MEASURED FOR EACH PROBE WITH AN AGILENT PRECISION SOURCE (SMU). IT IS EVIDENT THAT THERE IS NO SIGNIFICANT CHANGE IN THE VOLTAGE AND CURRENT MEASUREMENTS FOR EACH EXPOSURE TIME.

Probe	Time (h) ±1/10	Test 1 (mV) ±0,015%	Test 1 (mA) ±0,02%	Test 2 (mV) ±0,015%	Test 2 (mA) ±0,002%
1	24	4,9957	2,69984	5,0000	2,72986
2		4,9986	2,62312	5,0000	2,70863
3		4,9961	2,68015	5,0000	2,73036
4		4,9959	2,66441	5,0000	2,73050
5	48	4,9969	2,69269	5,0000	2,73237
6		4,9945	2,70077	5,0000	2,69253
7		4,9947	2,69642	5,0000	2,73996
8		4,9981	2,66461	5,0000	2,71784
9	72	4,9957	2,69916	5,0000	2,73399
10		4,9968	2,69147	5,0000	2,69172
11		4,9958	2,63288	5,0000	2,71659
12		4,9956	2,69116	5,0000	2,70748
13	120	4,9950	2,67401	5,0000	2,69887
14		4,9962	2,71427	5,0000	2,70543
15		4,9964	2,62276	5,0000	2,68232
16		4,9953	2,66035	5,0000	2,70193
17	144	4,9964	2,62358	5,0000	2,65368
18		4,9962	2,67669	5,0000	2,72137
19		4,9967	2,69907	5,0000	2,70022
21	168	4,9990	2,50334	5,0000	2,52335
22		4,9990	2,56698	5,0000	2,66136
23		4,9958	2,65978	5,0000	2,62487
24		4,9988	2,57720	5,0000	2,70076

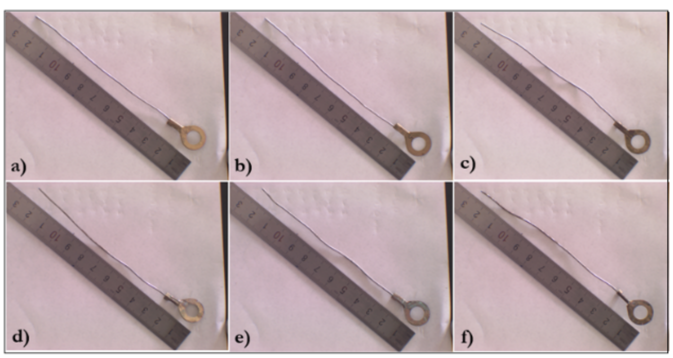

Figure 2. Photographs of probes at different hours of exposure in saline chamber. a) probe at 24 hours, b) probe at 48 hours, c) probe at 72 hours, d) probe at 120 hours, e) probe at 144 hours, f) probe at 168 hours. Is evident the absence of a determining corrosive effect [3].

Once the EC sensor was built, tested and validated, [6], [7], [8], [9] were considered as references for the use of Internet of Things (IoT) as a technological platform implemented with commercial off the shelf (COTS) components, to capture and analyze data for decision-making regarding climate change effects in greenhouse horticulture, and new sources of energy as pellets out of a microalgae culture.

III. AN EC SENSOR FOR HIGHLY SALINE MEDIUM

The authors then approached the design of an EC sensor for highly saline aqueous medium, to test this capability all probes were analyzed electrically and physically, both tests were approved without problems, see again table I. The final product is a system affordable for producers with limited economical means, and yet accurate enough for a researcher still exploring for alternative techniques of production. This requirements not only dictated the choosing of a COTS based system, but also promoted the use of online processing tools, that are commonly available at a low cost on the cloud. The result is a system under a thousand American dollars (with all sensors and connectivity included), but with a precision comparable to laboratory measurement equipment, and with data post processing available on the cloud.

A. Current measurement in each probe

In order to determine if after exposure to the saline chamber the electrodes underwent a change in their electrical characteristics, the measurement of the electrical current in milliamps (mA) was performed by means of a precision source at a fixed voltage of five (5) millivolts (mV). The precision source used was Agilent B2902A model; within its most relevant characteristics are: minimum measurement resolution is 100fA/100nV; minimum resolution of the source is 1pA/1micro V. This source is recommended because its precise characterization of the "Devices under test" (DUT), in the figure 3 is shown a schematic representation of the circuit developed for the measurements of the electrical current in each electrode at a fixed input voltage of 5V with an Agilent B2902A SMU.

IV. FUZZY LOGIC CONTROLLER

The output of the FLC gives a scalar value that is considered as a suggested value for irrigation decision-making. For example, see figure 4, for an output value of 0,367 in low irrigation

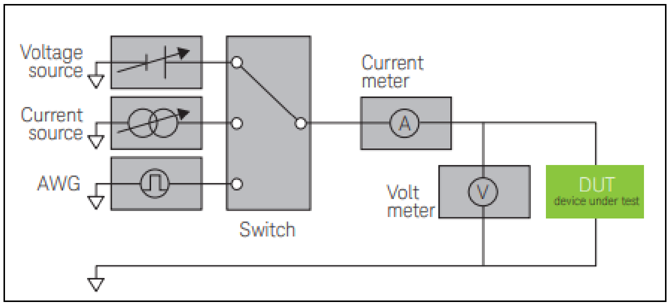

Figure 3. Standalone controller with IoT capabilities, automatic measurement of drainage, pH, EC, ambient temperature and humidity. Collected data is send to ThingSpeak IoT platform [10].

and 0,133 in normal irrigation, the final irrigation decision should be 13,3 % of normal irrigation value, because water is easily available, CE is good and pH is ideal. With a numerical value as output the lexical uncertainty is easily solved with the FLC. See again figure 4 with FLC results using QtFuzzyLite software version 6.0, the FuzzyLite Libraries for Fuzzy Logic Control [11]

A. *Embedded Fuzzy Logic Library*

In order to have a single hardware platform that allows decision-making, the embedded Fuzzy Logic Library (eFLL) [12] was selected to be included into the embedded system. It should be noted that other options were evaluated, for example the use of the Fuzzy Logic Library provided by MATLAB that can be used inside the ThingSpeak platform, but this alternative had an important limitation which is the dependency in a proprietary software and costs associated with the purchase of licenses for its use and update.

Another aspect taken into consideration for the development of the controller inside the embedded system and not inside the IoT platform, was the dependency with the provider of ThingSpeak platform, as well as the possibility of losing important data in case of failures in the remote system. Due to the above, a ”standalone” FLC was designed, it can take decisions on site and send the variables that were measured, filtered, optimized and analyzed by the controller to the IoT platform. Besides that, user have the possibility to acquire data locally or remotely.

The eFFL library is written in C ++/C, and only requires the standard C language library ”stdlib.h”, so eFLL is a library designed not only for Arduino controller, but any embedded system that is written with C commands.

Among the determining technical characteristics to select the use of eFLL are the following:

- It has no explicit limitations on quantity of fuzzy rules.
- There are no limitations on the number of inputs.
- There are no limitations on the number of outputs.
- Processing and storage limitations are associated only with the selected microcontroller.
- Contemplate the use of Maximum and Minimum (MAX-MIN) for processing.

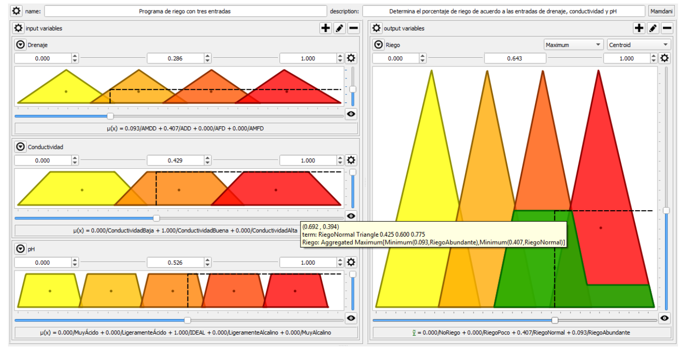

Figure 4. Irrigation decision considering drainage (water difficult available), EC (good EC) and pH (ideal pH), output (normal irrigation). QtFuzzyLite 6.0 FLC was configured with membership function and FL rules [11].

- Has a Mamdani controller integrated.
- Supports the use of triangular and trapezoidal shapes in both inputs and outputs.
- Has been tested with GTest for C by Google Inc.
- Free license use from MIT (Massachusetts Institute of Technology).
- Documentation updated and free to use.

In figure 5 is shown the final irrigation decision in the ThingSpeak platform, data is sent by the standalone FLC controller. This results are applicable to microalgae cultivation process, as the ones mentioned in [13], [14]. In [15], [16] some examples of crop water requirements and irrigation scheduling with conventional process are mentioned, proposed standalone FLC is also applicable to replace conventional techniques for decision-making.

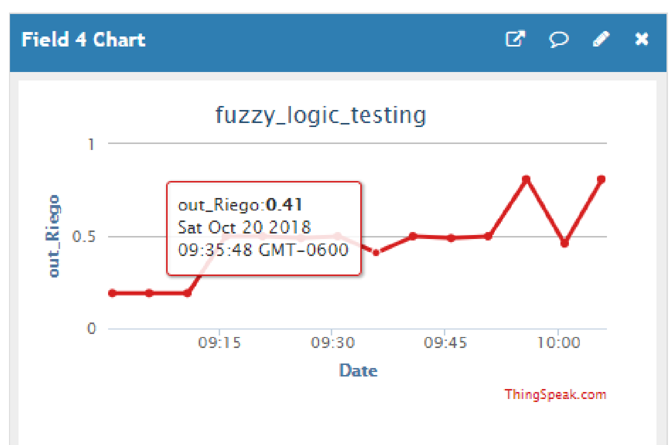

Figure 5. Final irrigation decision given for the FLC, automatic measurement of EC value is considered as a trigger for decision making. Collected data is send to ThingSpeak IoT platform [10].

Figure 6 shows an schematic representation of the EC sensor utilized as a trigger for decision-making in greenhouse horticulture measured drainage [3].

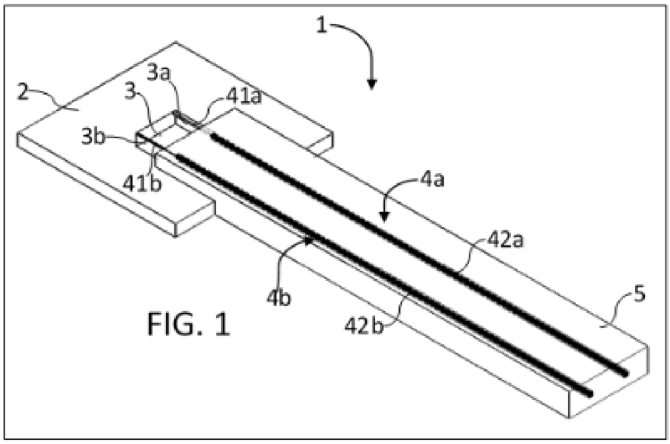

Figure 6. Final design of the EC sensor, patent pending documentation [3].

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V. Conclusions

A complete electronic embedded system, used to automatically enhance overdrain measurement accuracy in greenhouse horticulture, with IoT applications capable of storing and analyzing multiple measurement of environmental variables related to crop production in greenhouses, and including a standalone FLC for decision-making regarding irrigation has been presented.

A Fuzzy Logic Controller (FLC) for decision making about irrigation scheduling, was programmed using eFLL and is running in a standalone device, it was developed in an Arduino One microcontroller, that sends data to ThingSpeak IoT platform. The system was tested in a real greenhouse with vegetables crops. Researchers at Agricultural Engineering Department are collaborating with the post-processing data analysis in order to evaluate the algorithm's performance. The FLC output has been compared against QtFuzzyLite 6.0 output given value, results are consistent with expert criteria in both controllers.

All tests performed to the EC sensor validate that it is resistant to highly saline medium and that it can stay submerged in this medium without any calibration or maintenance, for long periods of time, which can be several months. Measurements with SMU also validate that there is not any electrical change in the EC sensor after 168 hours in salt spray chamber.

The developed system can be used for many other applications, such as: microalgae culture, forests and open air plantation monitoring, and even electrical smart grid monitoring. Only the sensors need be replaced for the intended application,

An affordable solution was thus implemented, which provides an integrated multivariable measurement environment, using COTS to trigger decision-making in greenhouse horticulture, in what is generally known as Internet of Things. and the data processing capabilities of the cloud can be used even for more specific applications, such as remote control and intelligent decision platforms.

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